

3.5 Derivative of Trigonometric Functions

Motivation:

Suppose we want to find the first derivative of $f(x) = \sin(x)$. If we use the definition of the derivative, we obtain:

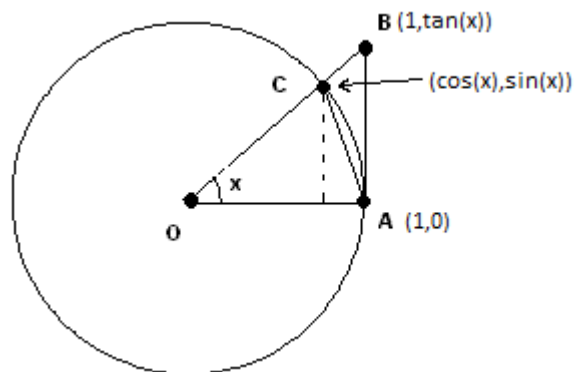
Theorem: If x is measured in radians, we have:

(1)

(2)

Proof (of 1):

We note that since x is approaching zero we may assume that x is the measure of an acute angle. Using the figure to the right, we compute 3 different areas:



Theorem: $\frac{d}{dx}[\sin(x)] = \cos(x)$

Proof:

Using the computation from the motivation we have that:

Example 1: Use the definition of the derivative to find $\frac{d}{dx}[\cos(x)]$.

(Hint: $\cos(A+B) = \cos(A)\cos(B) - \sin(A)\sin(B)$)

Example 2: Use the quotient rule to find the first derivative of each of the following.

a) $y = \sec(x)$

b) $y = \csc(x)$

c) $y = \cot(x)$

d) $y = \tan(x)$

Summary of Trig Function Derivatives:

$$\frac{d}{dx}[\sin(x)] = \cos(x)$$

$$\frac{d}{dx}[\cos(x)] = -\sin(x)$$

$$\frac{d}{dx}[\tan(x)] = \sec^2(x)$$

$$\frac{d}{dx}[\cot(x)] = -\csc^2(x)$$

$$\frac{d}{dx}[\sec(x)] = \sec(x)\tan(x)$$

$$\frac{d}{dx}[\csc(x)] = -\csc(x)\cot(x)$$

Example 3: Suppose that the position of an object on a vertical spring at time t is given by the function: $f(t) = 4\cos(t)$. Find a function for the velocity of the object at time t .

Example 4: Differentiate $f(x) = x^2 \sin(x) \cos(x)$.